

CHAPTER 32

Hearing aids for the treatment of tinnitus

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Abstract: Clinical evidence shows that the use of hearing aids in tinnitus patients provides two benefits: it makes the patient less aware of the tinnitus and it improves communication by reducing the annoying sensation that sounds and voices are masked by the tinnitus. Hearing loss reduces stimulation from external sounds resulting in increased awareness of tinnitus and deprivation of input may change the function of structures of the auditory pathways. Tinnitus is often caused by expression of neural plasticity evoked by deprivation of auditory input. With hearing aid amplification, external sounds can provide sufficient activation of the auditory nervous system to reduce the tinnitus perception and it may elicit expression of neural plasticity that can reprogram the auditory nervous system and thereby have a long-term beneficial effect on tinnitus by restoring neural function. To obtain the best results, hearing aids should be fitted to both ears, use an open ear aid with the widest amplification band, and disabled noise reducing controls. In some cases a combination device would be preferable. The conditions required in order to obtain good results include not only the use of devices, but above all, their adaptation to the needs of the single patient, by counseling and customization. Wearing the hearing aid must become second nature to the patient even though it is only one element of the therapy.

Keywords: hearing aid; sound therapy; sound generator; tinnitus; hearing loss; open ear device

Introduction

Epidemiological studies show that approximately 50% of individuals with hearing loss also have tinnitus (Davis, 1998). Other studies show other values, thus Sheldrake and Jastreboff (2004) showed that approximately 70% of individuals with hearing problems have tinnitus. The incidence of tinnitus is greater in the population that consults an ear nose and throat (ENT) specialist (Davis and Refaie, 2000). For this reason, the use

of hearing aids is justified as a means to reduce the effects of the tinnitus.

Saltzman and Ersner (1947) showed that patients with tinnitus benefited from using hearing aids. These results were subsequently confirmed by other studies (Kiessling, 1980; Stacey, 1980; Brooks and Bulmer, 1981; Miller, 1981; Melin et al., 1987). In literature it reported that approximately 50% of patients utilizing hearing aids experienced relief from the tinnitus (Surr et al., 1985). In later studies, Surr et al. (1999) reported an average improvement of approximately 10%, only 6 weeks after the hearing aid was fitted. On the other hand, Melin et al. (1987) concluded that hearing aids were not an effective means for

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reducing tinnitus. It must nevertheless be remembered that the hearing aids of that time were linear analog devices that were much less flexible in terms of their adaptability to the needs of the patient than modern digital hearing aids, and thus these old types of hearing aids were not suitable for patients with slight to medium levels of hearing loss or progressive hearing loss in the high-frequency range, which is often accompanied by tinnitus (Konig et al., 2006).

Folmer and Caroll (2006) compared 3 groups of 50 individuals with tinnitus using hearing aids, custom sound generator and other forms of acoustic therapy (music, relaxation CDs or the environmental sound machine). The participants that used hearing aids had hearing loss in the high-frequency range. After an average of 18 months after treatment, 70% of the participants who used hearing aids and 76% of those who used sound generator had improvements of their tinnitus. The participants in all groups reported a significant reduction ($p < 0.0001$) in their Tinnitus Severity Index (TSI) score.

Hearing aids or custom sound generators even reduced tinnitus in individuals with Ménière's disease (Herraiz et al., 2006a).

The authors own daily clinical experience — just like the experiences published (Sheldrake and Jastreboff, 2004) — has shown how the use of hearing aids provides two benefits: (1) it makes the patient less aware of the tinnitus, or even masks the tinnitus itself; (2) it improves communication and reduces the annoying sensation that sounds and voices are masked by the tinnitus.

Neurophysiological rationales for using hearing aids

The rationale behind the treatment of tinnitus by hearing aids is based on two complementary assumptions:

- (1) By increasing the level of ambient noise perceived by the patient, prosthetic amplification reduces or eliminates the contrast between endogenous sound (the tinnitus) and the silence caused by hearing loss (Frachet et al., 2004). This effect can be achieved by sound therapy, both in total

masking (Vernon and Meikle, 2000) or in partial masking of the tinnitus (mixing point) utilized in TRT, in accordance with Jastreboff's neurophysiological model of tinnitus (Jastreboff, 1999). The hearing aid is a useful instrument for habituating the patient to the perception of the noise of the tinnitus. In this way, the tinnitus gradually becomes less intrusive and thus less annoying: less attention is paid to the tinnitus, which is relegated to the category of minor stimuli, thus facilitating gradual habituation (Sheldrake and Jastreboff, 2004).

- (2) Most forms of tinnitus occur together with slight hearing loss, or hearing loss that is limited to the high frequencies (Konig et al., 2006). Hearing loss cause deprivation of input, which can lead to tinnitus through expression of neural plasticity (see Chapter 3) and it has been shown that it can lead to hyperactivity of the neurons in the auditory pathways (Kaltenbach, 2006) (see Chapter 9). Even a slight loss can cause changes in the function of parts of the auditory nervous system. Several studies have shown that changes caused by deprivation can be reversed by appropriate sound stimulation (Norena and Eggermont, 2006). Hearing aids are beneficial because they can restore auditory input and can stimulate cerebral plasticity (Gabriel et al., 2006). A mathematical model (Schaette and Kempter, 2006) has also been published, which predicts how tinnitus combined with neural hyperactivity could be reduced by means of adequate stimulation. The second rationale arises from this, maintaining that, thanks to the hearing aid, it is possible to stimulate cerebral plasticity and, at least, partly re-establish the proper functioning of the auditory nerve pathways, limiting one of the probable causes of tinnitus.

Hearing aids can facilitate the use of music in therapy of individuals with tinnitus and redirect attention toward complex and more interesting sounds (Searchfield, 2005) (see also Chapter 43).

The hearing aid as an instrument for the treatment of tinnitus

Whenever possible, the fitting should be done with open ear hearing aids. The importance of utilizing hearing aids with large ventilation ([Sheldrake and Jastreboff, 2004](#); [Searchfield, 2005](#)) has become more evident with the availability of open ear devices. The partial occlusion of the external auditory canal can lead to dissatisfaction with the use of the device and, in some cases, even lead to an increase in the perception of the tinnitus. The open fitting hearing aids also provide extended stimulation even at the higher frequencies, in the tinnitus pitch area. In this way, it is possible to obtain a dual result: providing the sound enrichment necessary for developing habituation to sound therapy and stimulating the neural plasticity of the auditory pathways, as discussed previously. The fitting of open ear devices in the treatment of tinnitus has been shown ([Del Bo et al., 2006](#)) to be as effective, if not more so, than for custom sound generators.

Whenever possible, hearing aids should be fitted in both ears since this allows better spatial localization and understanding of verbal messages. Both of which are important factors that make an additional contribution to controlling tinnitus by activating the entire auditory nervous system. In patients with deafness in one ear (anacusis), the use of a contralateral routing of signals (CROS) solution is advisable: the rationale for this approach is provided by the multisensorial stimulation of the auditory system, a situation that facilitates adaptation to the tinnitus ([Sheldrake and Jastreboff, 2004](#)).

Modern digital hearing aids are very versatile and equipped with numerous functions to improve hearing comfort, which are not, however, always suitable for the treatment of tinnitus. Directional microphones often reduce the amplification at low frequencies and thus it is advisable to activate the omnidirectional microphone; in addition, for the same reasons outlined above, it is appropriate to increase the amplification of weak, low-frequency sounds deactivating expansion and lowering compression activation point as well as disabling the digital noise reduction circuit. Obviously these

modifications must only be made if they are compatible with patient comfort and user satisfaction.

When utilizing open ear hearing aids, Digital Feedback Cancellation should be activated. It must be a contra phase control and not notch filter, so as to avoid a lessening of amplification at high frequencies. It is also appropriate to use hearing aids with the widest possible high-frequency amplification band because many patients have high-frequency hearing loss and high-frequencies amplification can provide stimulation in the auditory pathway in the area of tinnitus pitch. The tinnitus associated with profound deafness can be treated advantageously by means of a cochlear implant ([Matsushima et al., 1994](#); [Mo et al., 2002](#); [Rubinstein et al., 2003](#); [Jastreboff and Hazell, 2004](#)).

It has been estimated that 5% of patients who seek help for their tinnitus has normal hearing. For such patients custom sound generators are useful ([Searchfield et al., 2002](#)).

It is important always to recommend that the patient introduce environmental auditory enrichment, by using a table sound machine. Subjects with hearing loss and tinnitus can obtain an additional benefit from sound therapy by using a hearing aid with a built-in sound generator (combination instrument) ([Frachet et al., 2004](#)). In this case, the amplification of environmental signals is enriched by the addition of continuous stimulation from the hearing aid, with wide band sounds at a controlled level. In patients affected by transmission deafness following malformation of the outer and middle ear, use of a traditional hearing aid with custom ear mould is not advisable. According to [Holgers and Hakansson \(2002\)](#), 35% of these patients have tinnitus. In the aforementioned cases, the fitting of a bone-anchored hearing aid (BAHA) may be useful. Fitting the BAHA requires minor surgery that, in the adult patient, is normally done under local anesthesia ([Tjellstrom et al., 2001](#)). By using this type of hearing aid, we avoid any occlusion of the auditory canal and the patient is provided with a level of amplification suited to his level of hearing loss. Amplification can be used in the treatment of patients with tinnitus, in accordance with what has already been stated for traditional hearing aids.

The conditions required in order to obtain good results with hearing aid treatment for tinnitus include not only the use of technologically advanced devices, but above all, their adaptation to the needs of the patient, as well as counseling. The patient must be listened to, guided and kept informed, right from the diagnostic stage, throughout the setting up of the therapy and during the course of the adjustment and follow-up visits (Herraiz et al., 2006a, b). This will allow the therapist to understand the problems and attempt to resolve them, as far as possible, by appropriate fitting of the devices. This task involves all the technical and medical staff, which must always be able to respond to the patient's many requests for information with precision and in a coordinated manner.

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